Lab Exercise 19- ClusterIP in Service in Kubernetes

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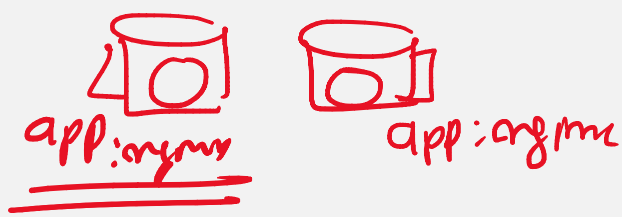
A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

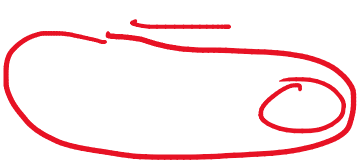
Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
```



HPA

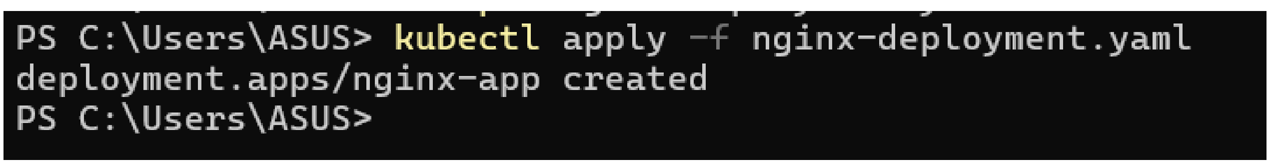


```
image: nginx
ports:
- containerPort: 80
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Save file.

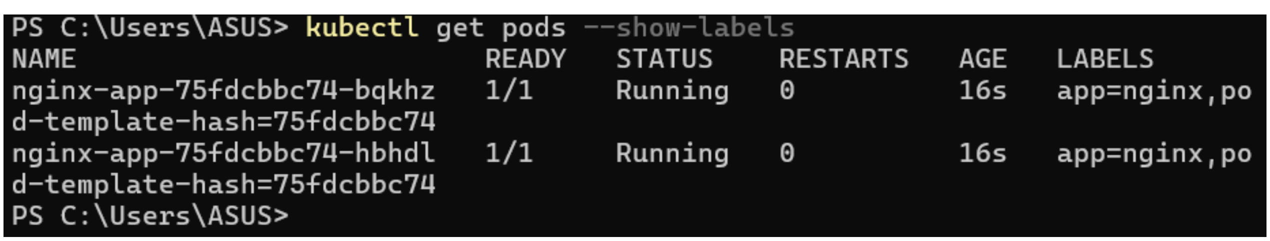
```
kubectl apply -f nginx-deployment.yaml
```



```
PS C:\Users\ASUS> kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created
PS C:\Users\ASUS>
```

Verify:

```
kubectl get pods --show-labels
```



```
PS C:\Users\ASUS> kubectl get pods --show-labels
NAME                         READY   STATUS    RESTARTS   AGE   LABELS
nginx-app-75fdcbbc74-bqkhz   1/1     Running   0          16s   app=nginx,po
d-template-hash=75fdcbbc74
nginx-app-75fdcbbc74-hbhdl   1/1     Running   0          16s   app=nginx,po
d-template-hash=75fdcbbc74
PS C:\Users\ASUS>
```

You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```


```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```
PS C:\Users\ASUS> kubectl apply -f nginx-service.yaml
service/nginx-service created
PS C:\Users\ASUS>
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

```
PS C:\Users\ASUS> kubectl get svc
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)
 AGE
kubernetes          ClusterIP   10.96.0.1        <none>        443/TCP
 11d
nginx-service       ClusterIP   10.109.211.174   <none>        80/TCP
 36s
nodeport-service    NodePort    10.108.246.39    <none>        80:30007/TCP
 11d
PS C:\Users\ASUS>
```

You should see:

```
TYPE       CLUSTER-IP
ClusterIP   10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

```
PS C:\Users\ASUS> kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 Endpo
intSlice
NAME            ENDPOINTS                        AGE
nginx-service   10.244.0.65:80,10.244.0.66:80    53s
PS C:\Users\ASUS>
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

```
PS C:\Users\ASUS> kubectl run test-pod --image=busybox --rm -it -- sh
All commands and output from this session will be recorded in container logs
, including credentials and sensitive information passed through the command
 prompt.
If you don't see a command prompt, try pressing enter.
/ #
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

```
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ #
```

Exit:

```
exit
```

```
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' comm
and when the pod is running
pod "test-pod" deleted from default namespace
PS C:\Users\ASUS>
```

```
PS C:\Users\ASUS> kubectl delete deployment nginx-app
deployment.apps "nginx-app" deleted from default namespace
PS C:\Users\ASUS> kubectl delete service nginx-service
service "nginx-service" deleted from default namespace
PS C:\Users\ASUS>
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.